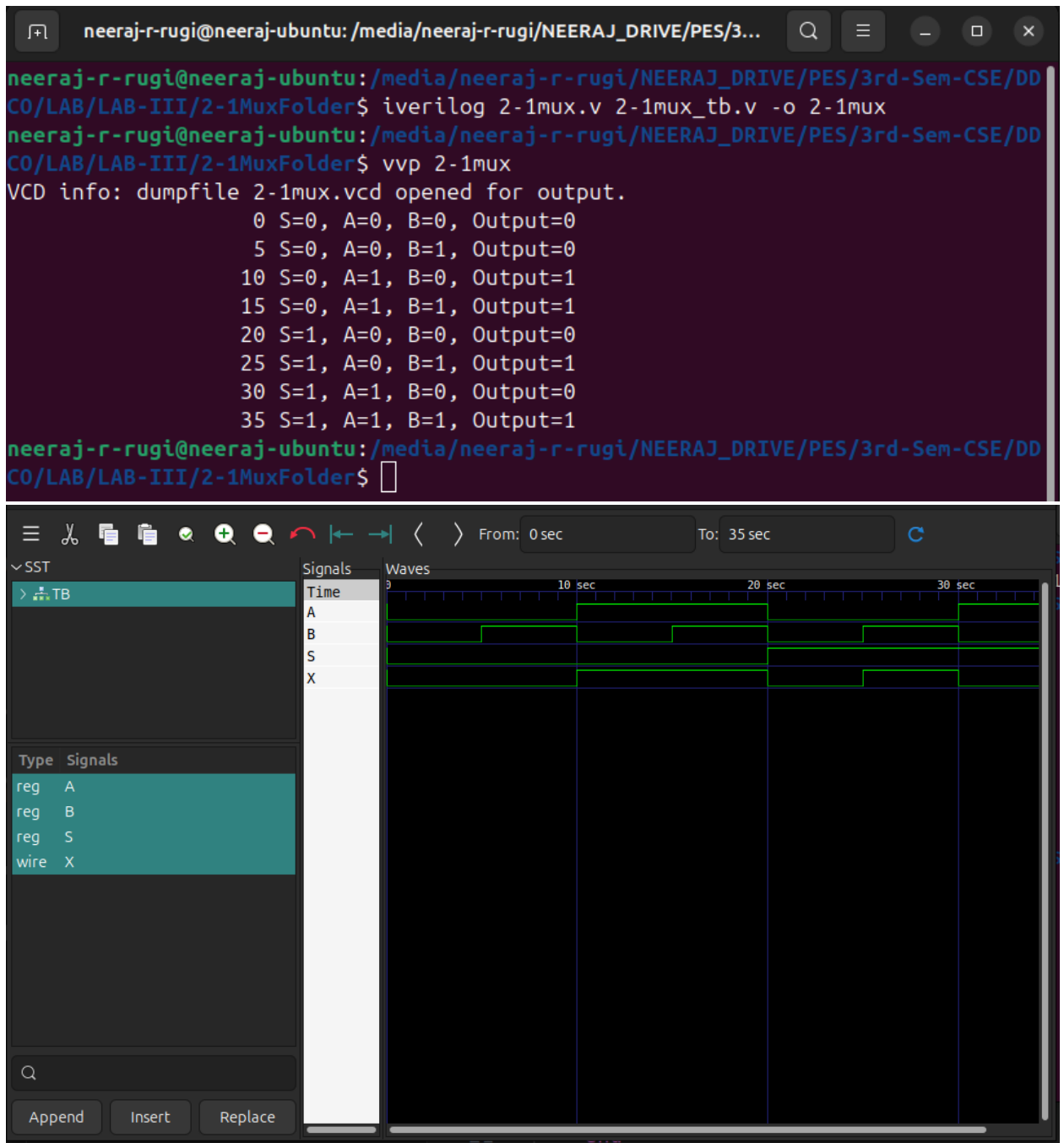


LAB - IV
Section: F
Part - I
SRN: PES2UG24CS311
Name: NEERAJ R RUGI

1. 2:1 Multiplexer

```
module mux2 (input wire i0, i1, j, output wire o);
    assign o = j==0?i0:i1;
endmodule

muxFolder > 2-1mux_tb.v
1  module TB;
2  reg A,B,S;
3  wire X;
4  initial
5      begin
6          $dumpfile("2-1mux.vcd");
7          $dumpvars(0,TB);
8      end
9  mux2 newMUX(.i0(A), .i1(B), .j(S), .o(X));
10 initial
11     begin
12         $monitor($time, " S=%b, A=%b, B=%b, Output=%b", S, A, B, X);
13         S = 1'b0;    A = 1'b0; B = 1'b0;
14         #5 S = 1'b0; A = 1'b0; B = 1'b1;
15         #5 S = 1'b0; A = 1'b1; B = 1'b0;
16         #5 S = 1'b0; A = 1'b1; B = 1'b1;
17         #5 S = 1'b1; A = 1'b0; B = 1'b0;
18         #5 S = 1'b1; A = 1'b0; B = 1'b1;
19         #5 S = 1'b1; A = 1'b1; B = 1'b0;
20         #5 S=1'b1; A = 1'b1; B = 1'b1;
21     end
22 endmodule
```



2.4:1 MUX

```

module mux2 (input wire i0, i1, j, output wire o);
    assign o = j==0?i0:i1;
endmodule

module mux4 (input wire [0:3] i, input wire j1, j0, output wire o);
    wire t0, t1;
    mux2 mux2_0 (i[0],i[1],j0,t0);
    mux2 mux2_1 (i[2],i[3],j0,t1);
    mux2 mux2_2 (t0,t1,j1,o);
endmodule

```

```

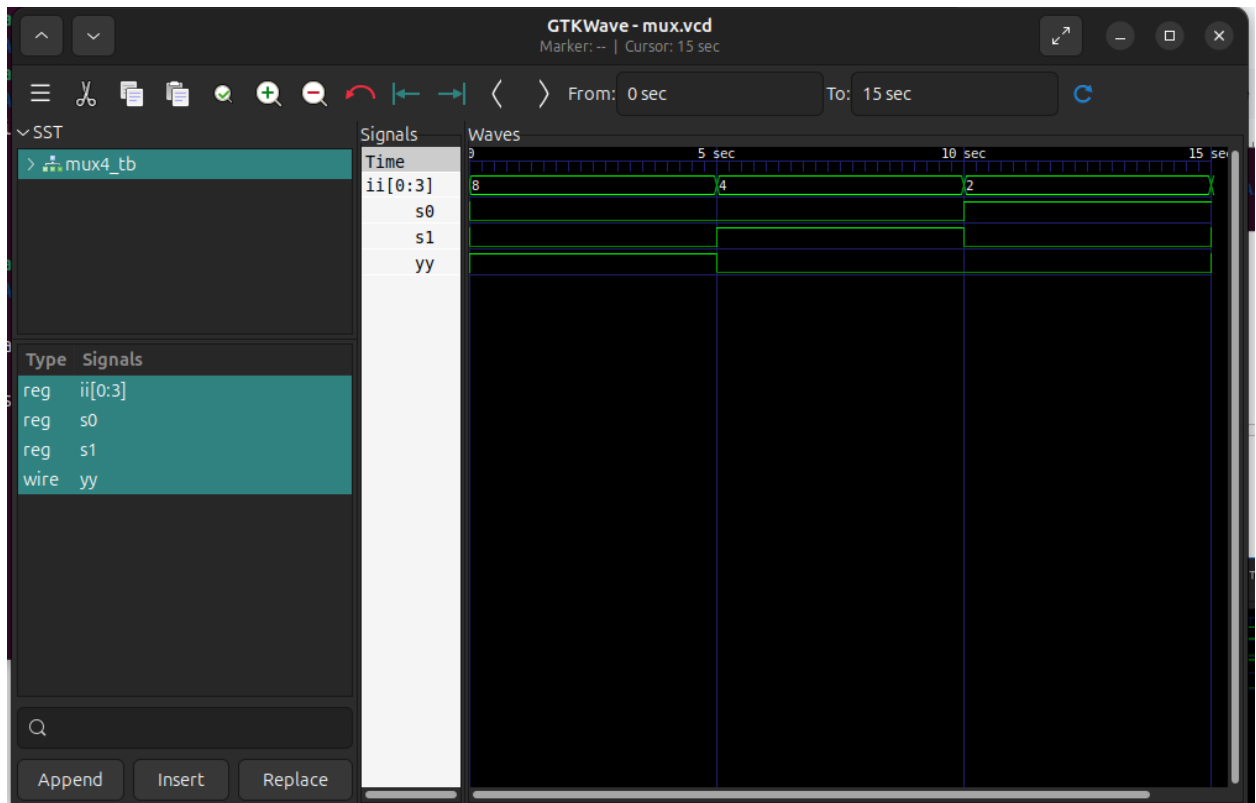
module mux4_tb;
    reg [0:3]ii;
    reg s0;reg s1;
    wire yy;
    initial
    begin
        $dumpfile("mux.vcd");
        $dumpvars(0, mux4_tb);
    end
    mux4 newMUX(.i(ii), .j0(s0),.j1(s1),.o(yy));
    initial
    begin
        $monitor($time, " ii = %b, j0 = %b, j1 = %b, o = %b", ii,s0,s1,yy);
        #5      ii = 4'b1000;s0=1'b0;s1=0'b0;
        #5      ii = 4'b0100;s0=1'b0;s1=1'b1;
        #5      ii = 4'b0010;s0=1'b1;s1=0'b0;
        #5      ii = 4'b0001;s0=1'b1;s1=1'b1;
    end
endmodule

```

```

neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/4-1MuxFolder$ iverilog 4-1mux.v 4-1mux_tb.v -o 4-1mux
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/4-1MuxFolder$ vvp 4-1mux
VCD info: dumpfile mux.vcd opened for output.
      0 ii = 1000, j0 = 0, j1 = 0, o = 1
      5 ii = 0100, j0 = 0, j1 = 1, o = 0
     10 ii = 0010, j0 = 1, j1 = 0, o = 0
     15 ii = 0001, j0 = 1, j1 = 1, o = 1
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/4-1MuxFolder$ 

```



3. Decoder

```
module decoder (input wire [1:0] a, input wire en, output wire [3:0] y);  
    assign y[3] = en & ~a[0] & ~a[1]; // * MSB  
    assign y[2] = en & a[0] & ~a[1];  
    assign y[1] = en & ~a[0] & a[1];  
    assign y[0] = en & a[0] & a[1]; // * LSB  
endmodule
```

```

module decoder_tb;
    reg [1:0] A;
    reg EN;
    wire [3:0] Y;
    initial
    begin
        $dumpfile("2-4decoder.vcd");
        $dumpvars(0, decoder_tb);
        $monitor("Time = %0t EN = %b A = %b Y = %b", $time, EN, A, Y);
    end
    decoder newDEC(.a(A), .en(EN), .y(Y));
    initial
    begin
        EN = 1'b0; A = 2'b00; #5;
        EN = 1'b1; A = 2'b00; #5;
        EN = 1'b1; A = 2'b01; #5;
        EN = 1'b1; A = 2'b10; #5;
        EN = 1'b1; A = 2'b11; #5;
        EN = 1'b0; A = 2'b10; #5;
    end
endmodule

```

```

neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/2-4Decoderfolder$ vvp 2-4decoder
VCD info: dumpfile 2-4decoder.vcd opened for output.
Time = 0 EN = 0 A = 00 Y = 0000
Time = 5 EN = 1 A = 00 Y = 1000
Time = 10 EN = 1 A = 01 Y = 0100
Time = 15 EN = 1 A = 10 Y = 0010
Time = 20 EN = 1 A = 11 Y = 0001
Time = 25 EN = 0 A = 10 Y = 0000
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/2-4Decoderfolder$ █

```

